

Lab03

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Lab03

```
library(readr)
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
getStats <- function(input, pos) {
  resultado <- input |>
  filter(!startsWith(DESTINATION_AIRPORT, "1")) |>
  filter(!is.na(DESTINATION_AIRPORT), !is.na(ARRIVAL_DELAY)) |>
  group_by(DESTINATION_AIRPORT) |>
  summarise(
    soma_dos_atrasos = sum(ARRIVAL_DELAY),
    total_voos = n(),
    .groups = "drop"
  )
  return(resultado)
}
```

```
airlines <- read_csv("airlines.csv")
```

Rows: 14 Columns: 2

-- Column specification -----

Delimiter: ","

chr (2): IATA_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
cols_airports <- cols_only(  
  IATA_CODE = col_character(),  
  CITY = col_character(),  
  STATE = col_character(),  
  LATITUDE = col_double(),  
  LONGITUDE = col_double()  
)  
airports <- read_csv("airports.csv", col_types = cols_airports)
```

```
dados_processados <- read_csv_chunked(  
  file = "flights.csv",  
  callback = DataFrameCallback$new(getStats),  
  chunk_size = 1000000,  
  col_types = cols_only(DESTINATION_AIRPORT = col_character(),  
                        ARRIVAL_DELAY = col_double())  
)
```

```
flights <- dados_processados |>  
  group_by(DESTINATION_AIRPORT) |>  
  summarise(  
    atraso_total = sum(soma_dos_atrasos),  
    volume_voos = sum(total_voos),  
    .groups = "drop"  
  ) |>  
  mutate(atraso_medio = atraso_total / volume_voos) |>  
  select(DESTINATION_AIRPORT, atraso_medio)
```

```
aeroportos_ausentes <- anti_join(  
  flights,  
  airports,
```

```

    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  )
print(aeropertos_ausentes)

```

```

# A tibble: 0 x 2
# i 2 variables: DESTINATION_AIRPORT <chr>, atraso_medio <dbl>

```

```

flights <- left_join(
  flights,
  airports,
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")
)

```

```

aeropertos_por_estado <- airports |>
  group_by(STATE) |>
  summarise(total_aeropertos = n()) |>
  arrange(desc(total_aeropertos))

aeropertos_por_estado

```

```

# A tibble: 54 x 2
  STATE total_aeropertos
  <chr>          <int>
1 TX              24
2 CA              22
3 AK              19
4 FL              17
5 MI              15
6 NY              14
7 CO              10
8 MN               8
9 MT               8
10 NC              8
# i 44 more rows

```

```

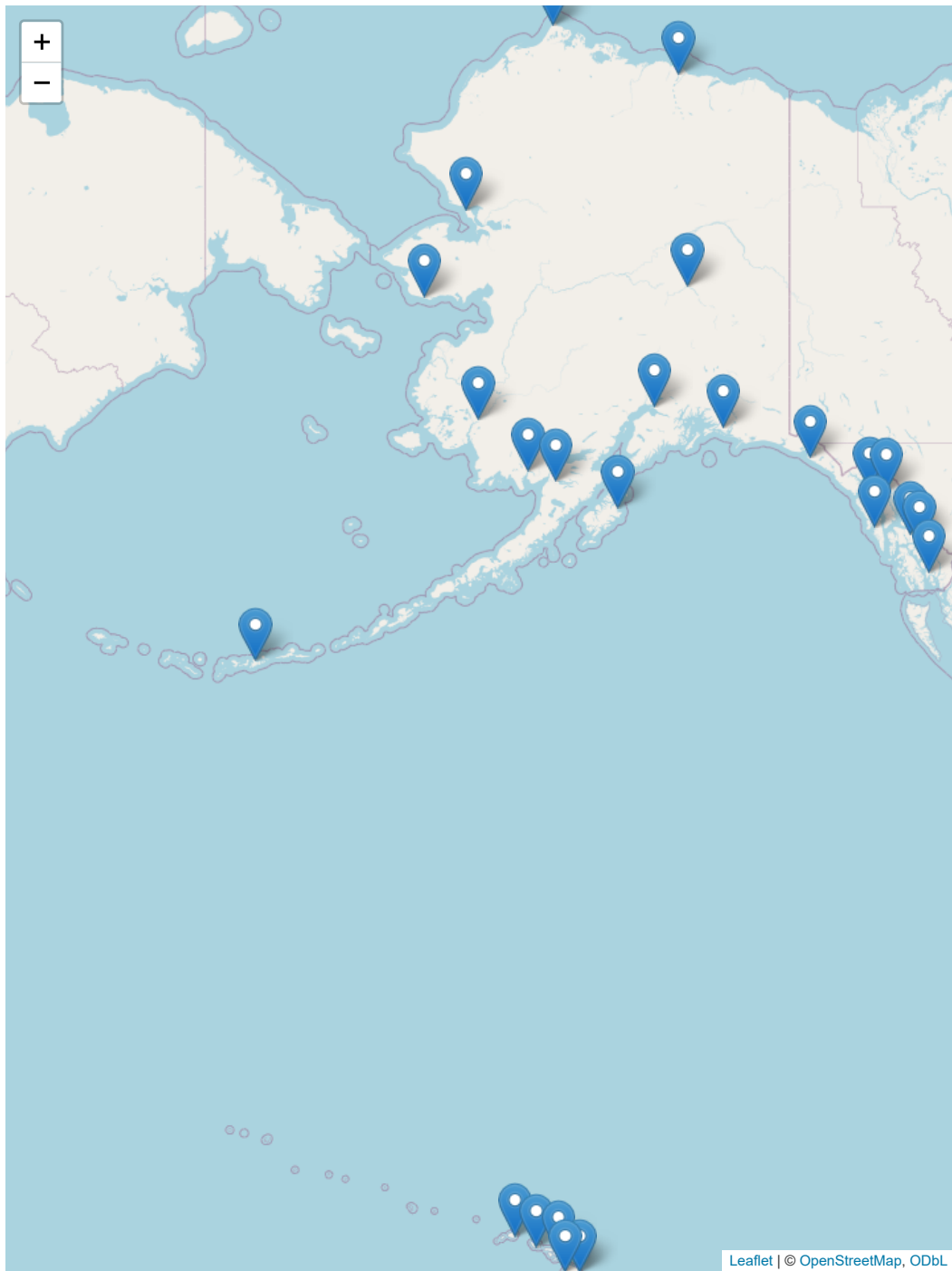
library(leaflet)

flights_mapa <- flights |>
  filter(!is.na(LATITUDE), !is.na(LONGITUDE))

```

```
this <- leaflet(data = flights_mapa) |>
addTiles() |>
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
  )
this
```

file:///C:/Users/afran/AppData/Local/Temp/RtmpqStpsF/file120c21129a3/widget120c304172e8.html



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-01 01:33:00 -03"
```